

Report 1 - Gradient Descent

Le Nghiem Thanh Thuy

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1 Algorithm

This algorithm uses gradient descent to minimize the function $y=x^2$. It starts from an initial value of x and updates the value step by step. In each iteration, x is adjusted using the gradient $2x$. The learning rate controls how big each step is. Over time, the value of x gets closer to 0, where the function reaches its minimum. Here the result with initial value 10 and learning rate 0.01

Table 1: Result table by time

Time	x	$f(x)$
1	8.00	64.00
2	6.40	40.96
3	5.12	26.21
4	4.10	16.78
5	3.28	10.74
6	2.62	6.87
7	2.10	4.40
8	1.68	2.81
9	1.34	1.80
10	1.07	1.15

2 Effect of learning rates

The table below shows the results of running the test with different learning rates.

Time	lr=0.01	lr=0.1	lr=0.2	lr=0.4
1	x=9.80, f=96.04	x=8.00, f=64.00	x=6.00, f=36.00	x=2.00, f=4.00
2	x=9.60, f=92.24	x=6.40, f=40.96	x=3.60, f=12.96	x=0.40, f=0.16
3	x=9.41, f=88.58	x=5.12, f=26.21	x=2.16, f=4.67	x=0.08, f=0.01
4	x=9.22, f=85.08	x=4.10, f=16.78	x=1.30, f=1.68	x=0.02, f=0.00
5	x=9.04, f=81.71	x=3.28, f=10.74	x=0.78, f=0.60	x=0.00, f=0.00
6	x=8.86, f=78.47	x=2.62, f=6.87	x=0.47, f=0.22	x=0.00, f=0.00
7	x=8.68, f=75.36	x=2.10, f=4.40	x=0.28, f=0.08	x=0.00, f=0.00
8	x=8.51, f=72.38	x=1.68, f=2.81	x=0.17, f=0.03	x=0.00, f=0.00
9	x=8.34, f=69.51	x=1.34, f=1.80	x=0.10, f=0.01	x=0.00, f=0.00
10	x=8.17, f=66.76	x=1.07, f=1.15	x=0.06, f=0.00	x=0.00, f=0.00

Figure 1: Compare result by learning rate

Here are the comments.

Table 2: Result table by time

Time	$lr = 0.01$		$lr = 0.1$		$lr = 0.2$		$lr = 0.4$	
	x	f(x)	x	f(x)	x	f(x)	x	f(x)
1	9.80	96.04	8.00	64.00	6.00	36.00	2.00	4.00
2	9.60	92.24	6.40	40.96	3.60	12.96	0.40	0.16
3	9.41	88.58	5.12	26.21	2.16	4.67	0.08	0.01
4	9.22	85.08	4.10	16.78	1.30	1.68	0.02	0.00
5	9.04	81.71	3.28	10.74	0.78	0.60	0.00	0.00
6	8.86	78.47	2.62	6.87	0.47	0.22	0.00	0.00
7	8.68	75.36	2.10	4.40	0.28	0.08	0.00	0.00
8	8.51	72.38	1.68	2.81	0.17	0.03	0.00	0.00
9	8.34	69.51	1.34	1.80	0.10	0.01	0.00	0.00
10	8.17	66.76	1.07	1.15	0.06	0.00	0.00	0.00

- A small learning rate leads to slow but stable convergence, often requiring many iterations.
- A large learning rate can speed up training but may cause overshooting or divergence.
- Choosing an appropriate learning rate is crucial for balancing convergence speed and stability.